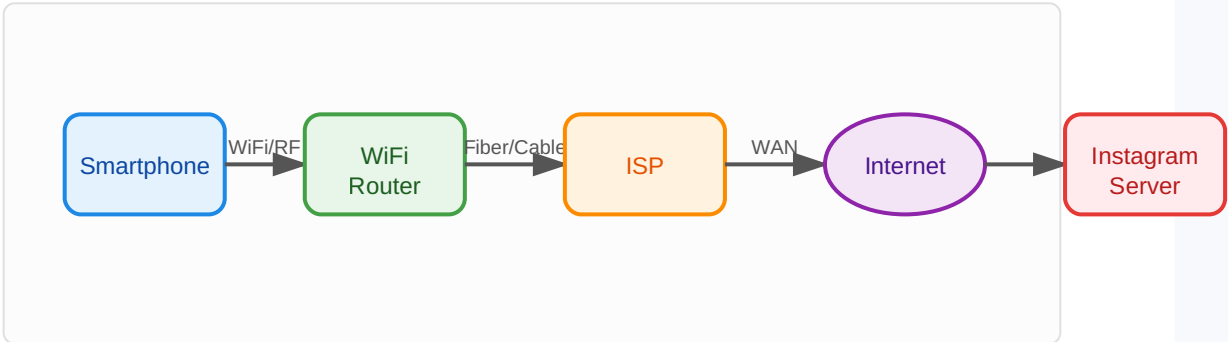


Computer Networks Basics

1. Connection Diagram: Phone to Instagram

The diagram below illustrates the typical path data takes from a personal smartphone to reach Instagram's servers.



2. Networked Devices in Daily Life

Here are 5 common devices and the types of networks they connect to:

Device	Network Type	Description
Smartphone	WLAN / WAN / PAN	Connects to home WiFi (WLAN), cellular networks (WAN), and Bluetooth accessories (PAN).
Laptop	WLAN / LAN	Connects to local WiFi networks (WLAN) or via an Ethernet cable (LAN) for internet access.
Smart TV	WLAN	Connects to the home wireless router (WLAN) to stream content from the internet.
Smartwatch	PAN	Connects directly to a smartphone via Bluetooth, forming a Personal Area Network (PAN).

Device	Network Type	Description
Wireless Router	LAN to WAN	Acts as the bridge connecting the local home network (LAN) to the wider internet (WAN).

3. Internet vs. Intranet

The primary difference between the Internet and an intranet lies in accessibility and scale:

- **Internet:** A global, public network of networks that anyone can access. For example, when you connect to a *public WiFi network at a cafe*, you are using the Internet to access universally available websites like Google or Wikipedia.
- **Intranet:** A private, restricted network contained within a specific organization. For example, a *college campus portal* used to check grades or access internal library systems might only be accessible when your device is connected to the college's internal campus WiFi. If you try to open that specific portal from your home network, it will not load because it is restricted to the college's intranet.

4. Data Travel: Performing a Search

App Chosen: Zomato

When I hit 'search' for a restaurant in Zomato, my smartphone sends the search query as a data packet over my local WiFi network (WLAN) to my home wireless router. The router forwards this packet through my Internet Service Provider (ISP) across the broader internet (WAN) until it reaches Zomato's web servers. Zomato's servers process my request, locate the restaurant data, and send the response packets back across the WAN to my ISP, through my router, and finally to my device, displaying the results on my screen.